

Technology Stack

Date	26 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS	Provides a simple and responsive user interface for entering weather parameters and displaying flood prediction results.

2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, loads the trained machine learning model, processes user input, and returns prediction results efficiently.
3	Database / Data Storage	Excel Dataset (flood_data.csv), Joblib	Historical flood data is stored in an Excel dataset. The trained model (best_model.pkl) and scaler (scaler.pkl) are stored using Joblib. No database is used in the current project.
4	Cloud / Hosting / Deployment	Localhost (Current), Render	The application currently runs on localhost. It can be deployed on Render.
5	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Joblib	These libraries are used for data preprocessing, visualization, model training, evaluation, and flood prediction.
6	Version Control & Source Code Management	Git, GitHub	Git is used for version control to track project changes, while GitHub is used to host the repository, manage source code, collaborate with team members, and maintain the project history.

